Predicting Family Income Using PSID (2013 & 2023)

I am using PSID Family Files from 2013 and 2023 to build a predictive model for family income

I am also looking to see what changes

Setup and Imports

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_squared_error, r2_score
```

Load PSID Family Files (2013 and 2023)

In [2]:
```
source_psid13 = pd.read_csv("psid_2013.csv")
source_psid23 = pd.read_csv("psid_2023.csv")
```

Renaming PSID Variables and Adding Year Column

In [3]:
```
psid13 = source_psid13.rename(columns={
    "ER53002": "id",
    "ER53016": "fam_size",
    "ER53017": "age",
    "ER53018": "sex",
    "ER53023": "marital",
    "ER53029": "own_status",
    "ER54306": "spouse_inc_flag",
    "ER55244": "health",
    "ER57839": "hours_worked",
    "ER58152": "fam_income",
    "ER58223": "education_years"
})
psid13["survey_year"] = 2013

psid23 = source_psid23.rename(columns={
    "ER82002": "id",
    "ER82017": "fam_size",
    "ER82018": "age",
    "ER82019": "sex",
    "ER82026": "marital",
```

```
    "ER82032": "own_status",
    "ER83494": "spouse_inc_flag",
    "ER84520": "health",
    "ER85326": "hours_worked",
    "ER85629": "fam_income",
    "ER85780": "education_years"
})
psid23["survey_year"] = 2023

cols = [
    "id", "survey_year", "fam_size", "age", "sex", "marital",
    "own_status", "spouse_inc_flag", "health",
    "hours_worked", "fam_income", "education_years"
]

psid13 = psid13[cols]
psid23 = psid23[cols]
```

Combine 2013 and 2023 into a Single Dataset

In [4]:
```
psid = pd.concat([psid13, psid23], ignore_index=True)
psid.head(), psid["survey_year"].value_counts()
```

Out[4]:
```
(       id  survey_year  fam_size   age  sex  marital  own_status  \
 0  7517.0         2013       1.0  59.0  2.0      4.0         1.0
 1  8684.0         2013       2.0  55.0  2.0      4.0         5.0
 2  8434.0         2013       2.0  37.0  1.0      2.0         5.0
 3  8569.0         2013       1.0  25.0  2.0      2.0         8.0
 4  2176.0         2013       3.0  28.0  2.0      5.0         5.0

    spouse_inc_flag  health  hours_worked  fam_income  education_years
 0              0.0     3.0          50.0    170000.0             14.0
 1              0.0     4.0           0.0     55080.0             11.0
 2              0.0     2.0           0.0      7000.0             10.0
 3              0.0     3.0          60.0      3450.0             12.0
 4              0.0     2.0          40.0     44340.0              9.0  ,
 survey_year
 2023    12661
 2013     9063
 Name: count, dtype: int64)
```

Cleaning and Creating Derived Variables

Convert PSID missing codes to NaN

For variables like fam_size, age, and health I converted them to numeric and replaced PSID missing codes with NaN. This allows pandas and scikit-learn to properly handle missing values when I later filter or drop them

New Binary Variables

sex_male - 1 if male, 0 if female

spouse_inc_yes - 1 if spouse had income, 0 if not

own_own - 1 if own home, 0 otherwise

own_rent - 1 if rent, 0 otherwise

year_2023 - 1 for 2023 observations, 0 for 2013

Also had to get rid of all negative values in fam_income so I could log transform, making the distribution more sutible for regression. Showing the difference between the raw data and the log transformation

In [5]:
```
psid = psid.copy()

bad_codes = {8, 9, 98, 99, 998, 999, 9998, 9999}

for col in ["fam_size", "age", "marital", "own_status",
            "spouse_inc_flag", "health", "hours_worked",
            "fam_income", "education_years"]:
    psid[col] = pd.to_numeric(psid[col], errors="coerce")
    psid.loc[psid[col].isin(bad_codes), col] = np.nan


psid["sex_male"] = np.where(psid["sex"] == 1, 1,
                      np.where(psid["sex"] == 2, 0, np.nan))


psid["spouse_inc_yes"] = np.where(psid["spouse_inc_flag"] == 1, 1,
                           np.where(psid["spouse_inc_flag"] == 5, 0, np.nan))


psid["own_own"] = np.where(psid["own_status"] == 1, 1, 0)
psid["own_rent"] = np.where(psid["own_status"] == 5, 1, 0)


psid["year_2023"] = (psid["survey_year"] == 2023).astype(int)

psid = psid[psid["fam_income"] > 0].copy()
psid["log_income"] = np.log(psid["fam_income"])

psid[["fam_income", "log_income"]].describe()
```

Out[5]:

	fam_income	log_income
count	1.781900e+04	17819.000000
mean	8.274441e+04	10.823910
std	1.289215e+05	1.102488
min	3.000000e+00	1.098612
25%	2.820000e+04	10.247077
50%	5.628000e+04	10.938095
75%	1.024250e+05	11.536886
max	6.301000e+06	15.656219

Summary Statistics for new dataset

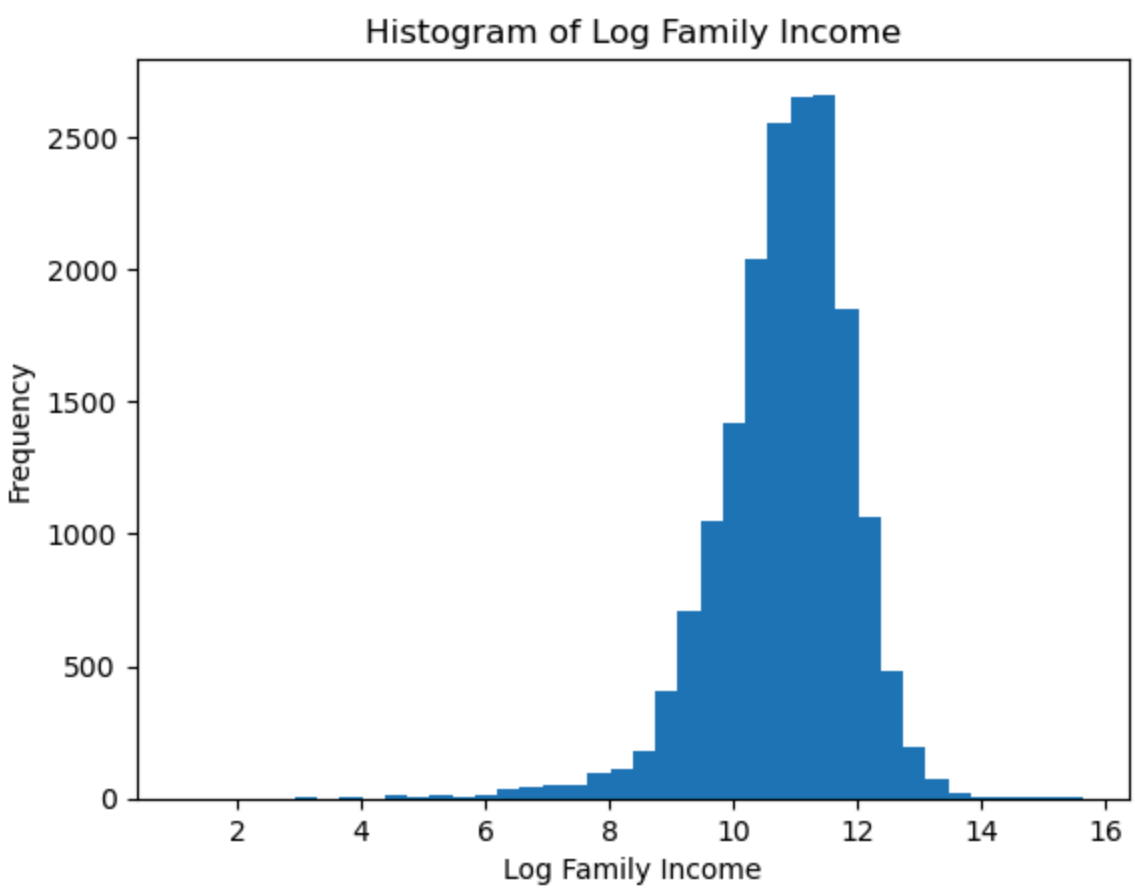
In [6]:
```
psid.describe()
```

Out[6]:

	id	survey_year	fam_size	age	sex	marital	own_sta
count	17819.000000	17819.000000	17748.000000	17813.000000	17819.000000	17817.000000	16850.000
mean	4560.378529	2017.970537	2.555217	46.645427	1.324317	2.022338	2.756
std	2642.218785	5.000053	1.445638	16.484773	0.468132	1.208810	1.985
min	1.000000	2013.000000	1.000000	17.000000	1.000000	1.000000	1.000
25%	2269.000000	2013.000000	1.000000	33.000000	1.000000	1.000000	1.000
50%	4559.000000	2013.000000	2.000000	44.000000	1.000000	2.000000	1.000
75%	6846.000000	2023.000000	3.000000	59.000000	2.000000	2.000000	5.000
max	9189.000000	2023.000000	13.000000	101.000000	2.000000	5.000000	5.000

Histograms of Log Family Income

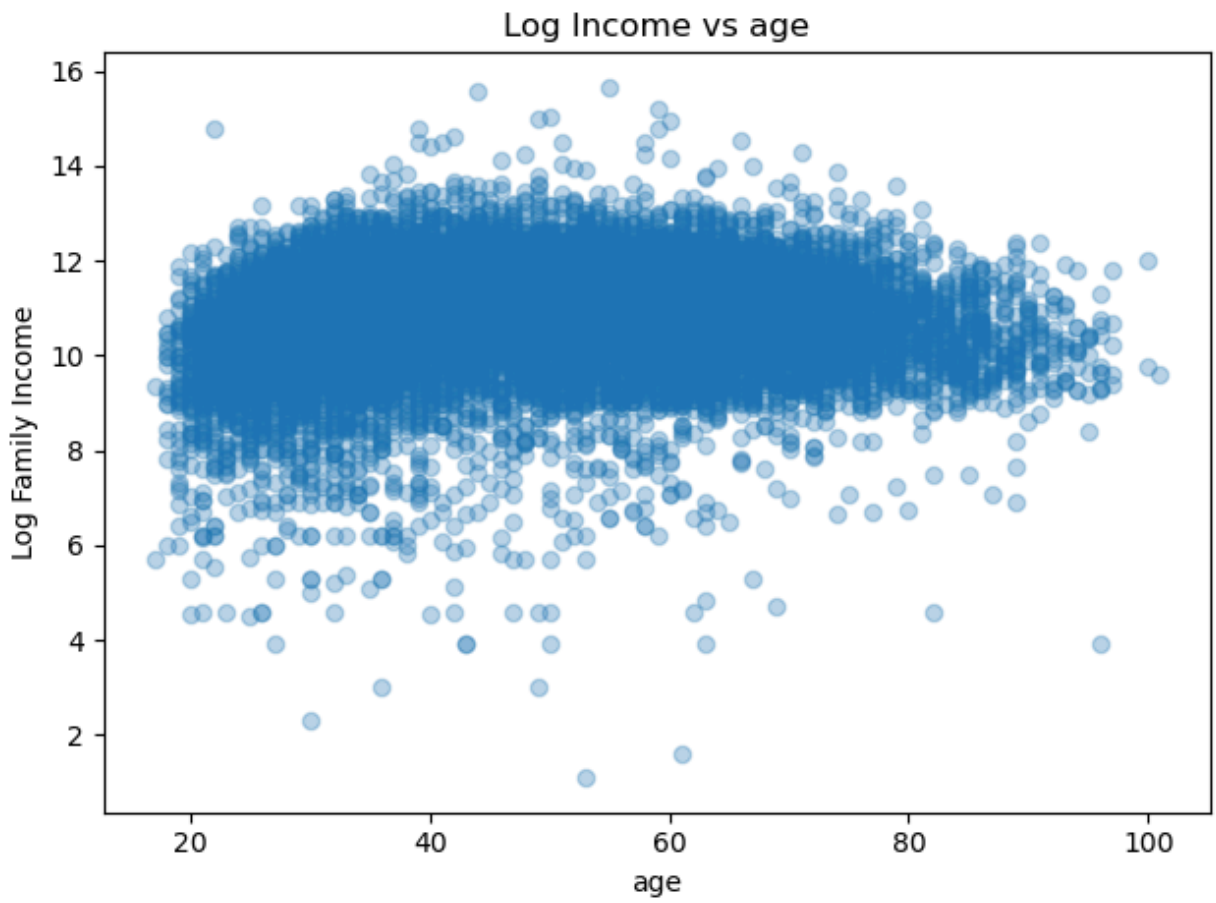
In [7]:
```
plt.hist(psid["log_income"], bins=40)
plt.xlabel("Log Family Income")
plt.ylabel("Frequency")
plt.title("Histogram of Log Family Income")
plt.show()
```

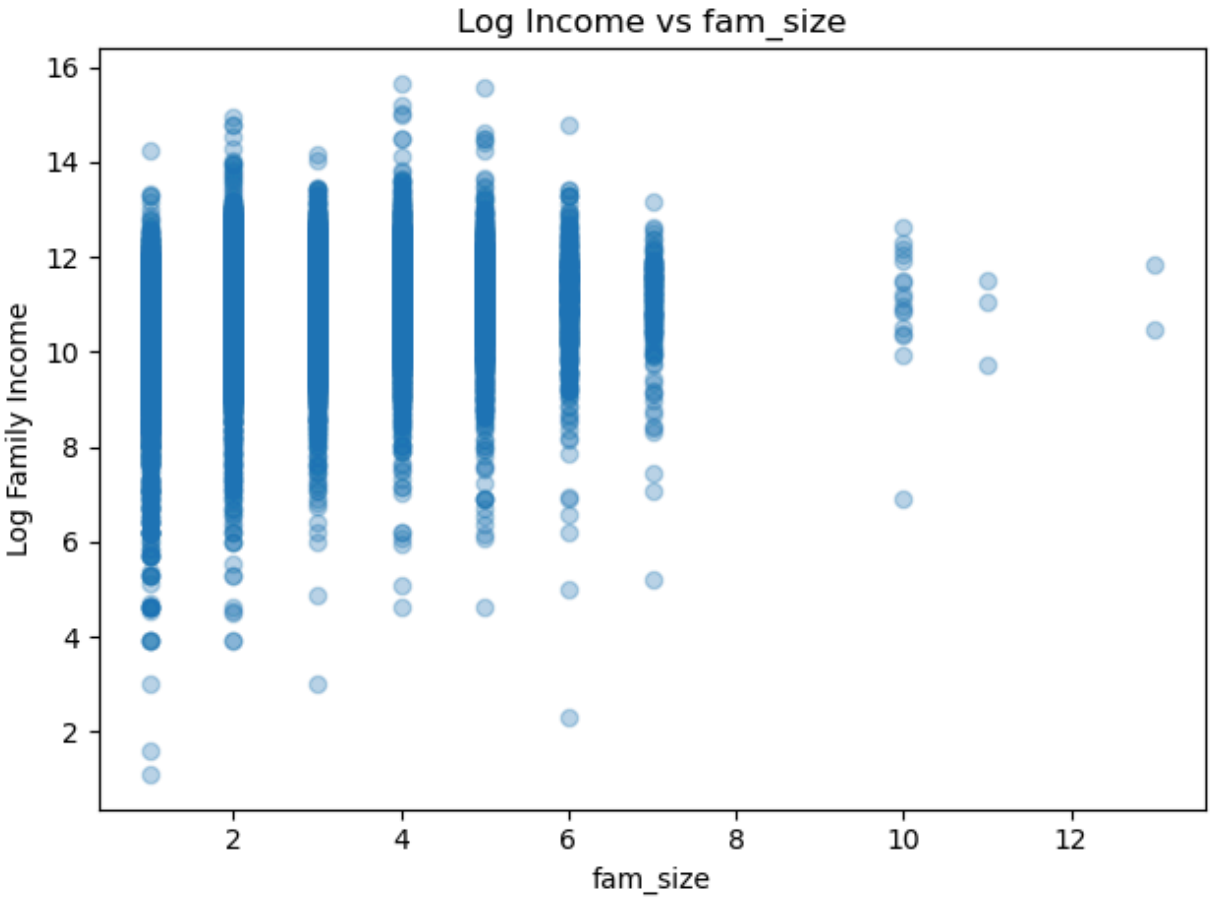


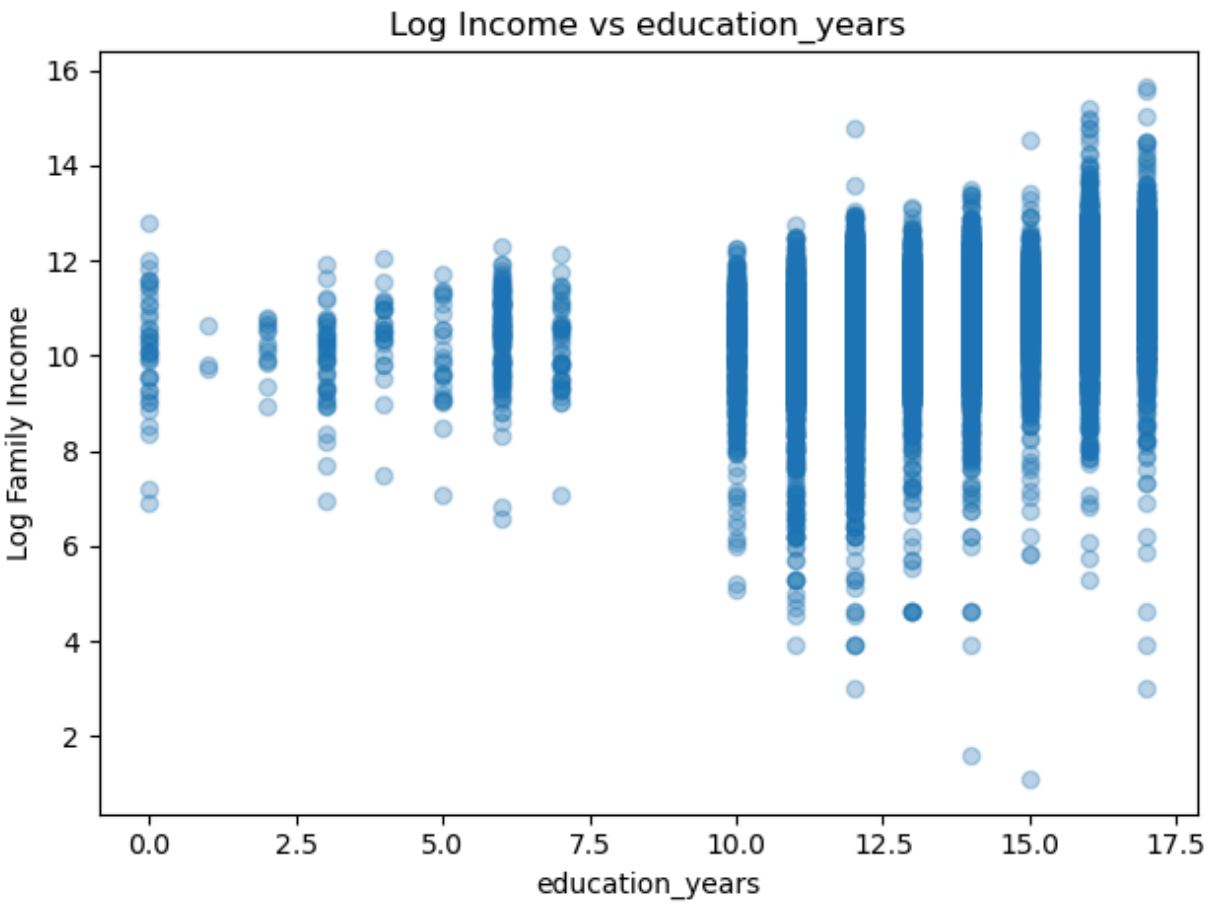
Scatterplots of Income vs Key Predictors by Year

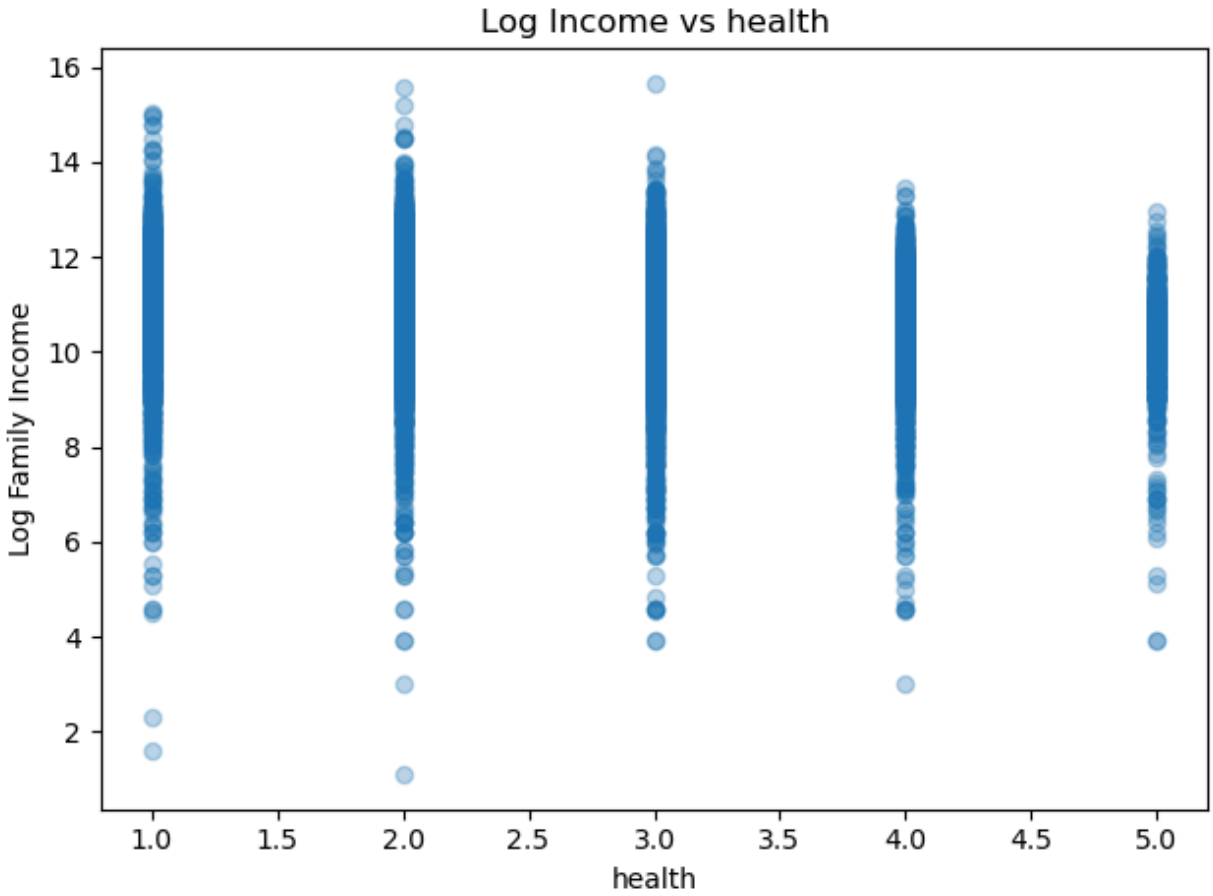
Plotting the scatterplots of log family income against non binary predictors

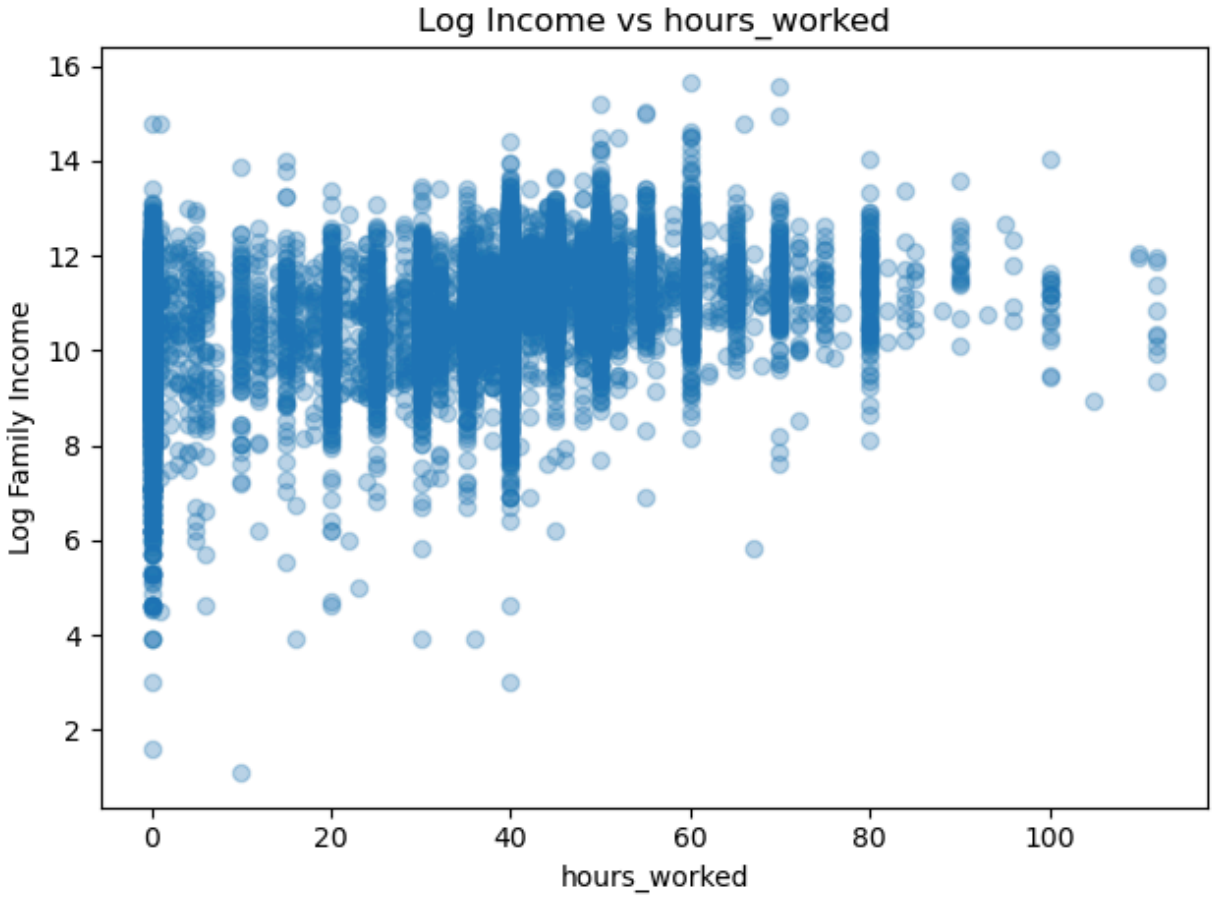
In [8]:
```
predictor_names = [
    "age", "fam_size", "education_years", "health",
    "hours_worked"
]

for var in predictor_names:
    plt.scatter(psid[var], psid["log_income"], alpha=0.3)
    plt.xlabel(var)
    plt.ylabel("Log Family Income")
    plt.title(f"Log Income vs {var}")
    plt.tight_layout()
    plt.show()
```




Log Income vs fam_size
Log Family Income
fam_size
16
14
12
10
8
6
4
2
2
4
6
8
10
12


Log Income vs education_years
Log Family Income
education_years
16
14
12
10
8
6
4
2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5





Building the Modeling Dataset and Train–Test Split

I dropped the rows with any missing values

I used my predictors as X and log_income as y and used an 80/20 split

In [9]:
```
model_data = psid.dropna(subset=[
    "log_income", "age", "fam_size", "education_years",
    "sex_male", "health", "hours_worked",
    "own_own", "own_rent", "spouse_inc_yes"
]).copy()

X = model_data[[
    "age", "fam_size", "education_years",
    "sex_male", "health", "hours_worked",
    "own_own", "own_rent", "spouse_inc_yes"
]]
y = model_data["log_income"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Linear Regression

I'm now fitting the model

I then calculate RMSE and R^2

In [10]:
```
linreg = LinearRegression()
linreg.fit(X_train, y_train)

y_pred_lr = linreg.predict(X_test)

lr_rmse = np.sqrt(mean_squared_error(y_test, y_pred_lr))
lr_r2 = r2_score(y_test, y_pred_lr)

print("Linear Regression RMSE:", lr_rmse)
print("Linear Regression R^2:", lr_r2)
```

```
Linear Regression RMSE: 0.6326892078882635
Linear Regression R^2: 0.34172351843780624
```

I ran an OLS summary to check if our variables were significant.

It showed the sex of the head person had a high p-value and was not statistically significant in predicting household income

In [11]:
```
X_sm = sm.add_constant(X)

ols_full = sm.OLS(y, X_sm).fit()
print(ols_full.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             log_income   R-squared:                       0.382
Model:                            OLS   Adj. R-squared:                  0.381
Method:                 Least Squares   F-statistic:                     586.0
Date:                Mon, 17 Nov 2025   Prob (F-statistic):               0.00
Time:                        23:17:55   Log-Likelihood:                -8291.4
No. Observations:                8556   AIC:                         1.660e+04
Df Residuals:                    8546   BIC:                         1.667e+04
Df Model:                           9
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const              8.2523      0.118     70.144      0.000       8.022       8.483
age                0.0108      0.001     18.221      0.000       0.010       0.012
fam_size           0.0322      0.006      5.573      0.000       0.021       0.044
education_years    0.1065      0.003     37.141      0.000       0.101       0.112
sex_male          -0.0257      0.087     -0.294      0.769      -0.197       0.146
health            -0.0476      0.008     -6.335      0.000      -0.062      -0.033
hours_worked       0.0129      0.000     33.044      0.000       0.012       0.014
own_own            0.5559      0.051     10.853      0.000       0.455       0.656
own_rent           0.2178      0.052      4.191      0.000       0.116       0.320
spouse_inc_yes     0.3595      0.016     21.975      0.000       0.327       0.392
==============================================================================
Omnibus:                      936.434   Durbin-Watson:                   1.747
Prob(Omnibus):                  0.000   Jarque-Bera (JB):             5729.973
Skew:                          -0.341   Prob(JB):                         0.00
Kurtosis:                       6.951   Cond. No.                     1.27e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spec
ified.
[2] The condition number is large, 1.27e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```

When using a KNN model you need to choose a value of k

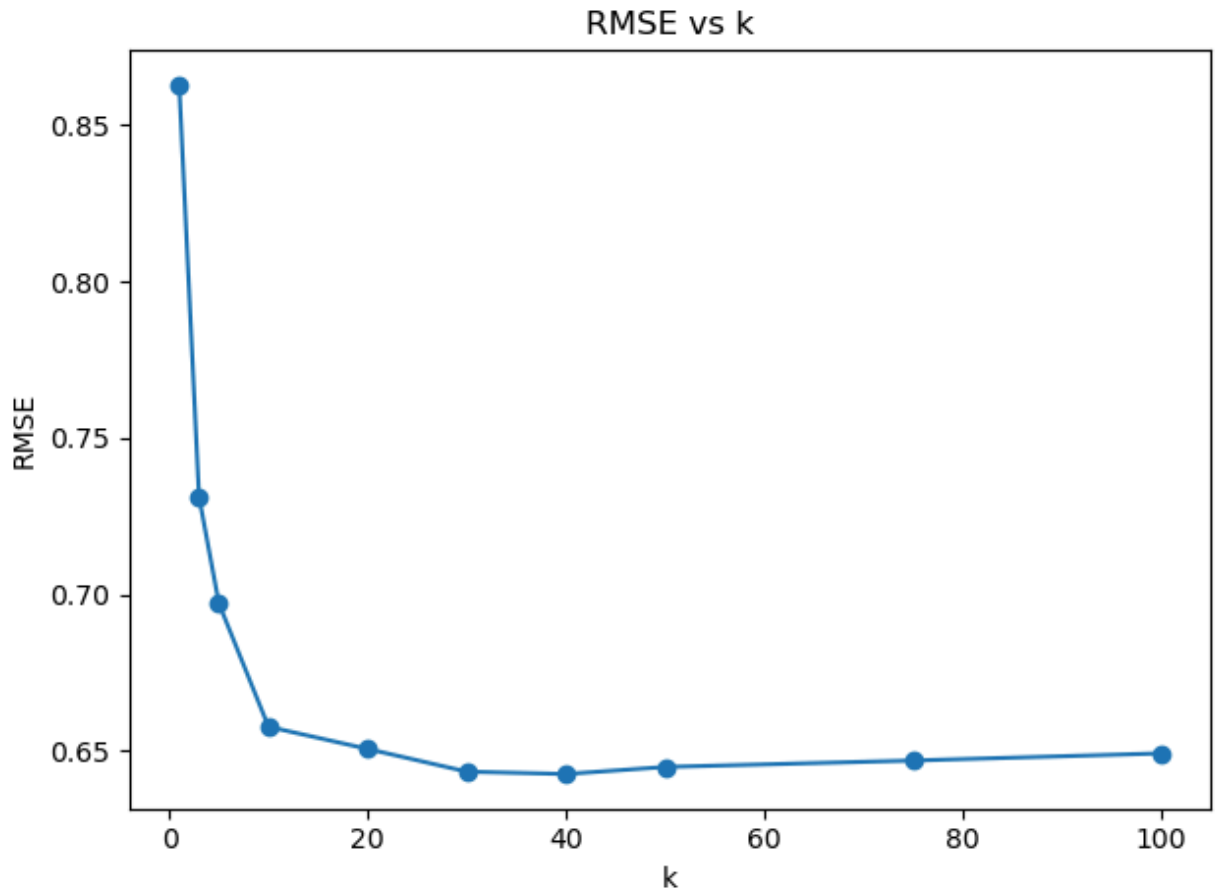
I used the elbow method

By looking at the graph, when k=30 that is when improvements in RMSE levels off

In [12]:

```
ks = [1,3,5,10,20,30,40,50,75,100]
rmses = []

for k in ks:
    knn = KNeighborsRegressor(n_neighbors=k)
    knn.fit(X_train, y_train)
    pred = knn.predict(X_test)
    rmse = np.sqrt(mean_squared_error(y_test, pred))
    rmses.append(rmse)

plt.plot(ks, rmses, marker='o')
plt.xlabel("k")
plt.ylabel("RMSE")
plt.title("RMSE vs k")
```

```
plt.tight_layout()
plt.show()
```



I fit the model and computed RMSE and R^2

In [13]:
```
knn_final = KNeighborsRegressor(n_neighbors=30)
knn_final.fit(X_train, y_train)

knn_pred = knn_final.predict(X_test)

knn_rmse = np.sqrt(mean_squared_error(y_test, knn_pred))
knn_r2 = r2_score(y_test, knn_pred)

print("KNN RMSE:", knn_rmse)
print("KNN R^2:", knn_r2)
```

```
KNN RMSE: 0.6433995439456835
KNN R^2: 0.31924790962665306
```

When looking at the RMSE and R^2 of our Linear Regression and KNN models, we see that they are similar, but the linear regression model performs better. A lower RMSE and higher R^2 is better.

In [14]:
```
results = pd.DataFrame({
    "Model": [
        "Linear Regression",
        "KNN"
    ],
    "RMSE": [
```

```
        lr_rmse,
        knn_rmse
    ],
    "R^2": [
        lr_r2,
        knn_r2
    ]
})

results.round(4)
```

Out[14]:

	Model	RMSE	R^2
0	Linear Regression	0.6327	0.3417
1	KNN	0.6434	0.3192

Because our linear regression model performed better I decided to run the same model, but split it by year, to see possible changes in our model's coeficicents, seeing how the same variables may change in influence over time.

We also see that our 2013 model performed better

In [15]:

```
X_cols = ["age","fam_size","education_years","sex_male","health",
          "hours_worked","own_own","own_rent","spouse_inc_yes"]

data13 = model_data[model_data["survey_year"] == 2013].dropna(subset=X_cols)
data23 = model_data[model_data["survey_year"] == 2023].dropna(subset=X_cols)

X13 = data13[X_cols]
y13 = data13["log_income"]

X23 = data23[X_cols]
y23 = data23["log_income"]

model13 = LinearRegression().fit(X13, y13)
model23 = LinearRegression().fit(X23, y23)

r2_2013 = model13.score(X13, y13)
r2_2023 = model23.score(X23, y23)

print("R^2 2013:", r2_2013)
print("R^2 2023:", r2_2023)
```

```
R^2 2013: 0.4233723648130573
R^2 2023: 0.39197279524488216
```

We can now look at the coefficients and see how things have changed from 2013 to 2023 in relation to family income.

In [16]:

```
coef_df = pd.DataFrame({
    "coef_2013": model13.coef_,
    "coef_2023": model23.coef_
}, index=X_cols)

coef_df.round(3)
```

Out[16]:

	coef_2013	coef_2023
age	0.012	0.009
fam_size	0.027	0.027
education_years	0.102	0.101
sex_male	0.000	0.186
health	-0.046	-0.080
hours_worked	0.014	0.012
own_own	0.659	0.464
own_rent	0.317	0.156
spouse_inc_yes	0.363	0.433

Age has had a slightly weaker effect over time

Family Size has had no change

Education has had basically no change

In 2013 sex made no difference, but in 2023 a man being the head person is associated with having a higher household income

Health had an even stronger negative association in 2023 compared to 2013

Hours worked was very similar

Both owning a home and renting both saw a steep fall

Having a spouse with income saw a significant rise from 2013 to 2023